

A SEMINAR REPORT ON CHAINED DEEP LEARNING MODEL FINE-GRAINED CYBERBULLYING DETECTION WITH BYSTANDER DYNAMICS

Submitted By

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AUGUST 2026
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In partial fulfillment of the requirements for the award of the degree
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C E R T I F I C A T E

This is to certify that, the seminar report titled ***A Chained Deep Learning Model for Fine-Grained Cyberbullying Detection With Bystander Dynamics*** is a bonafide record of the **CSQ413 SEMINAR** presented by **AMRUTHA G BABU(CEC23CS030)**, Seventh Semester B.Tech. Computer Science & Engineering student, under our guidance and supervision, in partial fulfillment of the requirements for the award of the degree B.Tech. Computer Science & Engineering of APJ Abdul Kalam Technological University.

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ACKNOWLEDGEMENT

This seminar work would not have been possible without the support and encouragement of many people. First and foremost, I give thanks to Almighty God who gave me the strength, resources and ability to complete the seminar successfully.

I would like to express my sincere gratitude to our Principal, **Dr. Jaya VL**, for providing the necessary facilities and support for the completion and presentation of this seminar. I am also deeply grateful to the Head of the Department, **Dr. Preetha Theresa Joy**, for her encouragement and support throughout the seminar work.

I extend my heartfelt thanks to my seminar co-ordinator, **Mrs. Deepa C G**, Assistant Professor, Department of Computer Science and Engineering, for her valuable coordination and guidance. I am especially grateful to my guide, **Mrs. Rohitha R**, Assistant Professor, Department of Computer Science and Engineering, for her constant guidance, encouragement and constructive suggestions throughout this seminar.

I would also like to thank my dear friends for their cooperation and encouragement throughout the seminar. Their support and understanding made the completion of this work easier and more meaningful.

ABSTRACT

Cyberbullying detection on social networking sites is challenging because harmful behaviour is often expressed implicitly and its meaning depends on the surrounding conversation. Conventional approaches frequently reduce the problem to binary classification of a single post, which may overlook conversational context and the influence of other participants. The paper reviewed in this seminar addresses this limitation through a chained deep learning model for fine-grained cyberbullying detection with bystander dynamics. The approach uses a session-based Twitter dataset, CYBY23, containing both bystander-role annotations and fine-grained cyberbullying labels. In the first stage, a fine-tuned BERT model identifies bystander roles from conversational information. The predicted role is then passed through a classifier-chain relationship to a second-stage LSTM model, which combines the textual representation with the predicted bystander information to classify the incident into bullying with high aggression, bullying with low aggression, or aggression without bullying. The reported implementation uses BERT-base-cased and three parallel LSTM branches followed by feature concatenation. The experimental results show that incorporating bystander-role information improves the weighted F1-score from 0.71 for the no-bystander baseline to 0.89, corresponding to a reported improvement of 25.35%. The study therefore demonstrates that modelling conversational participants and label dependency can improve fine-grained cyberbullying detection. The authors collected approximately 13,300 tweets between January 2022 and January 2023 and selected 150 threads containing approximately 1,024 tweets for annotation; after removing low-agreement data, 639 tweets remained, with a Fleiss' Kappa of 0.6078. The BERT stage uses a maximum sequence length of 256 tokens, a batch size of 16, an intermediate dense layer of 512 units, AdamW optimization with a learning rate of 1×10^{-5} and weight decay of 1×10^{-6} , and five fine-tuning epochs. The second stage uses three parallel LSTM branches for reply ID, tweet text and predicted bystander roles; experiments found 64 hidden units to provide the best overall trade-off, while dropout and early stopping further improved the configuration. The study also evaluates alternative BERT configurations, optimizers, epoch counts and recurrent architectures, providing a detailed experimental basis for the proposed chained model.

Contents

List of Figures	v
List of Tables	vi
1 Introduction	1
1.1 Paper Information	1
1.2 Background	2
1.3 Cyberbullying and Cyber-Aggression	2
1.4 Role of Bystanders	2
1.5 Need for Fine-Grained Detection	3
2 Fundamentals of Cyberbullying	4
2.1 Cyberbullying	4
2.2 Cyber-Aggression	4
2.3 Cyberbullying and Cyber-Aggression	4
2.4 Bystander Roles	5
2.5 Fine-Grained Detection	5
2.6 Classifier Chain	5
2.7 BERT and LSTM	5
3 Literature Survey	7
3.1 Binary and Feature-Based Detection	7
3.2 Session and Conversation-Based Detection	7
3.3 Participant-Role-Based Detection	7
3.4 Fine-Grained Severity Detection	8
3.5 Research Gap	8

4 Methodology	9
4.1 CYBY23 Dataset	9
4.2 Annotation Process	10
4.3 Bystander-Role Distribution	10
4.4 Classifier Chain	10
4.4.1 Classifier Chain Formulation	11
4.5 Overall Methodology	12
4.6 Layer 1: Bystander Role Model using BERT	13
4.7 BERT Bystander-Role Results	13
4.8 Layer 2: Fine-Grained Cyberbullying Model using LSTM	14
4.9 Two-Stage Classification and Chaining	15
4.10 Research Objectives and Contributions	15
4.11 Problem Formulation	15
4.12 Data Collection and Construction of CYBY23	16
4.13 Three-Level Annotation Strategy	17
4.14 Label Structure of the Dataset	17
4.15 Data Preparation and Training Split	18
4.16 Detailed BERT Architecture	18
4.17 BERT Input Configuration Experiments	19
4.18 Frozen BERT and RoBERTa Comparison	19
4.19 Detailed LSTM Architecture	20
4.20 LSTM Hyperparameter Experiments	20
4.21 Why the Chain Matters	21
4.22 Evaluation Metrics	21
5 Results and Discussion	23
5.1 Experimental Setup	23
5.2 Baseline Evaluation	23
5.3 Impact of Bystander Roles	23
5.4 Reliability of the Results	24

5.5 Comparison with Related Approaches	24
5.6 Theoretical and Practical Implications	25
5.7 Discussion	26
5.8 Analysis of the Baseline Comparison	26
5.9 Interpretation of the Confidence Interval	27
5.10 Experimental Findings from the Model Selection Process .	27
5.11 Error Propagation in the Chained Architecture	28
5.12 Comparison with Earlier Detection Strategies	28
5.13 Practical Significance	29
5.14 Overall Interpretation of the Results	29
6 Limitations and Future Scope	31
6.1 Limitations	31
6.2 Future Scope	31
7 Conclusion	33
References	34

List of Figures

4.1	Comparison of Binary Relevance and Classifier Chain problem transformation.	11
4.2	Two sequential layers of the chained fine-grained cyberbullying detection model.	12
4.3	Overall methodology from data collection and annotation to two-stage classification.	12
4.4	LSTM architecture used in the second stage.	14

List of Tables

4.1	Distribution of fine-grained cyberbullying classes in CYBY23.	9
4.2	Bystander-role detection results for the reported BERT configurations.	13
4.3	Detailed label structure used in the CYBY23-based chained classification task.	18
5.1	Impact of bystander-role information on final classification performance.	23
5.2	Confidence intervals for the proposed model and the baseline.	24
5.3	Comparison with selected related cyberbullying detection approaches.	24

Chapter 1

Introduction

1.1 Paper Information

The seminar is based on the research article *A Chained Deep Learning Model for Fine-Grained Cyberbullying Detection With Bystander Dynamics* by Haifa Saleh Alfurayj, Syaheerah Lebai Lutfi and Ramesh Perumal. The article was published in *IEEE Access*, Volume 12, in 2024, with article pages 105588–105604 and DOI 10.1109/ACCESS.2024.3435840. The paper was received on 3 July 2024, accepted on 23 July 2024, and published on 30 July 2024. The central contribution is a two-stage chained model in which bystander-role prediction is used as additional information for fine-grained cyberbullying classification.

Cyberbullying has become an important problem in social networking environments because harmful communication can spread rapidly through public and semi-public conversations. The paper reviewed in this seminar argues that cyberbullying language is often implicit and that the meaning, intention and severity of a message cannot always be determined by considering the main post alone. Instead, the broader conversation, including replies from other users, can provide important contextual information.

Traditional automated cyberbullying detection systems commonly formulate the task as a binary classification problem: bullying or not bullying. Such a formulation can be inadequate when aggressive communication does not satisfy all of the characteristics of cyberbullying. The distinction between cyber-aggression and cyberbullying therefore becomes important when designing a more precise detection system.

The work discussed in this seminar introduces a two-stage chained architecture. A BERT-based model is first used to identify bystander roles, and the predicted role is then supplied as additional information to an LSTM-based classifier for fine-grained cyberbullying classification. The classifier-chain formulation is used to preserve dependency between the two prediction tasks.

1.2 Background

Cyber-aggression can be viewed as aggressive behaviour in an online environment that is intended to cause harm. It may occur as an isolated incident and does not necessarily involve a persistent relationship or a power imbalance. Cyberbullying is more specific: it involves repetitive or frequent aggression and a power imbalance between the aggressor and the victim. These distinctions matter because a one-time aggressive message should not automatically be treated as a bullying episode.

The broader conversation is also important. In a social-media thread, responses from other participants may reinforce an aggressive post, oppose it, remain neutral, or be unrelated to the original topic. The paper therefore treats bystander behaviour as a useful contextual feature for fine-grained detection.

1.3 Cyberbullying and Cyber-Aggression

The two concepts can be differentiated using two principal characteristics:

1. **Frequency and repetition:** cyberbullying involves repeated aggression, whereas cyber-aggression may be a single incident.
2. **Power imbalance:** cyberbullying involves a power imbalance that makes it difficult for the victim to defend themselves; this is not required for cyber-aggression.

The paper emphasizes that confusing these categories can reduce the reliability of cyberbullying detection systems. Its fine-grained formulation therefore distinguishes bullying from aggression that does not qualify as bullying.

1.4 Role of Bystanders

Bystanders are participants in a conversation thread other than the bully and the victim. The CYBY23 annotation scheme used in the paper identifies four role categories:

- **Instigator:** agrees with or supports the author of the main post.
- **Defender:** disagrees with the thread topic and shows a defending response.
- **Impartial:** remains neutral or passive.
- **Other:** posts content that is unrelated to the thread topic.

The importance of these roles is that they may influence how a conversation develops. The paper therefore attempts to use bystander information not merely as descriptive metadata, but as an additional feature in the final cyberbullying classifier.

1.5 Need for Fine-Grained Detection

A binary decision of bullying versus non-bullying does not fully describe an aggressive interaction. The CYBY23 dataset used in the paper provides three fine-grained classes: bullying with high aggression, bullying with low aggression, and aggression without indication of bullying. This allows the final classifier to distinguish between aggressive content and actual bullying while also representing severity.

Chapter 2

Fundamentals of Cyberbullying

This chapter introduces the basic concepts required to understand the methodology discussed in the later chapters. It covers cyberbullying, cyber-aggression, bystander participation, fine-grained classification, and the basic ideas behind classifier chains, BERT and LSTM.

2.1 Cyberbullying

Cyberbullying refers to harmful or aggressive behaviour carried out through electronic communication and social networking platforms. A cyberbullying incident is not defined only by the presence of offensive language. Repetition and the presence of a power imbalance are important characteristics that distinguish bullying from a single aggressive interaction.

2.2 Cyber-Aggression

Cyber-aggression is broader than cyberbullying. It refers to aggressive online behaviour intended to cause harm, but it may occur as an isolated incident and does not necessarily involve a power imbalance. Thus, an aggressive message can exist without meeting the conditions required to classify the interaction as cyberbullying.

2.3 Cyberbullying and Cyber-Aggression

The distinction between the two concepts is mainly based on two factors:

1. **Frequency and repetition:** cyberbullying involves repeated aggression, whereas cyber-aggression may be a one-time incident.
2. **Power imbalance:** cyberbullying involves an imbalance that makes it difficult for the victim to defend themselves; cyber-aggression does not necessarily include this condition.

The reviewed paper uses this distinction when constructing its fine-grained classification task, allowing aggression without an indication of bullying to be treated separately from bullying with different aggression levels.

2.4 Bystander Roles

A bystander is a participant in a conversation thread other than the bully and the victim. The CYBY23 annotation scheme identifies four roles:

- **Instigator:** agrees with or supports the main post author.
- **Defender:** disagrees with the thread topic and responds in a defending manner.
- **Impartial:** remains neutral or passive.
- **Other:** posts content unrelated to the thread topic.

These roles provide additional social context that is not available when only the main post is considered.

2.5 Fine-Grained Detection

A binary classifier produces only two outcomes, such as bullying and not bullying. The reviewed approach instead distinguishes three final classes: bullying with high aggression, bullying with low aggression, and aggression without indication of bullying. Fine-grained classification therefore provides a more informative description of the incident.

2.6 Classifier Chain

A classifier chain is a multi-label problem-transformation method in which predictions are made sequentially. The prediction produced by one classifier is appended to the feature set used by the next classifier. In contrast, Binary Relevance trains an independent classifier for each label. [5]

2.7 BERT and LSTM

BERT (Bidirectional Encoder Representations from Transformers) is a transformer-based language model that learns contextual representations from text. In the reviewed system, a fine-tuned BERT model performs the first stage of classification by identifying bystander-role information.

LSTM (Long Short-Term Memory) is a recurrent neural network designed for sequential data. Its memory cell and gates allow useful information to be maintained across a sequence. In the reviewed system, LSTM forms the second

stage and uses the predicted bystander information together with textual inputs to perform fine-grained cyberbullying classification. [6]

Chapter 3

Literature Survey

Research in cyberbullying detection has evolved from traditional text classification toward contextual, session-based and transformer-driven approaches. The paper reviewed in this seminar summarizes several lines of work that motivate the proposed chained architecture.

3.1 Binary and Feature-Based Detection

Early and widely used approaches frame cyberbullying detection as binary classification. Deep learning models such as LSTM have been applied to text-based detection, while TF-IDF has been used to represent word-frequency information. More recent work uses transformer models such as BERT to capture contextual relationships in text. The paper also notes multimodal approaches that combine textual representations with image features.

3.2 Session and Conversation-Based Detection

A second line of research argues that an entire conversation thread is more informative than an isolated post. Ziems et al. incorporated contextual thread information and several cyberbullying criteria, including aggression, repetition, intent, peer visibility and power imbalance. Other work considered user interactions and temporal information through graph-based methods. These approaches show the value of context, but they do not explicitly combine bystander-role information with fine-grained bullying severity. [2]

3.3 Participant-Role-Based Detection

Some studies have attempted to identify participant roles such as harasser, victim and bystander. Jacobs et al. investigated automatic classification of participant roles in cyberbullying and demonstrated that social-media participants can be categorized using machine-learning and transformer-based methods. However, the data and task formulation differ from the session-based CYBY23 setting used in the reviewed paper. [3]

3.4 Fine-Grained Severity Detection

A smaller body of work considers cyberbullying severity rather than only a binary label. Wu and Tang proposed a hierarchical squashing-attention network for classifying cyberbullying severity in Chinese-language dialogues. This demonstrated that multi-class severity analysis is feasible, but it did not incorporate explicit bystander-role information. [4]

3.5 Research Gap

The paper identifies a gap at the intersection of two ideas: conversation-aware cyberbullying detection and bystander-role modelling. Existing studies may use session context, participant roles, or fine-grained severity labels, but the reviewed work specifically combines these aspects through a chained architecture. The key motivation is that bystander behaviour and bullying severity are dependent labels, so information predicted in the first stage can be useful for the second stage.

Chapter 4

Methodology

The methodology consists of data collection and annotation, preparation of the CYBY23 dataset, classifier-chain based processing, a BERT-based bystander-role classifier, and an LSTM-based fine-grained cyberbullying classifier. The overall objective is to allow information about bystander behaviour to influence the final cyberbullying decision.

4.1 CYBY23 Dataset

Approximately 13,300 tweets were collected over a twelve-month period from January 2022 to January 2023. Because complete manual annotation is time-consuming, the authors selected 150 tweet threads containing approximately 1,024 tweets for detailed annotation. The annotation process was carried out at tweet, reply and thread levels. Threads with low agreement were excluded, leaving 639 tweets with a Fleiss' Kappa value of 0.6078. [1]

The corpus contains two types of labels. The first is the fine-grained cyberbullying label: bullying with high aggression, bullying with low aggression, or aggression without indication of bullying. The second is the bystander-role label: instigator, defender, impartial, or other.

Table 4.1: Distribution of fine-grained cyberbullying classes in CYBY23.

Class	Percentage
Bullying with high aggression	11.61%
Bullying with low aggression	54.46%
Aggression without bullying	33.93%

The class distribution shows that the dataset is imbalanced, with high-aggression bullying representing the smallest class. This is an important consideration when interpreting the reported weighted metrics.

4.2 Annotation Process

Annotation was performed in three levels. At the tweet level, the main post was assigned an aggressiveness score using a three-point scale of 0, 1 and 2. At the replies level, the roles of bystanders were identified from the replies. At the thread level, the initial judgement could be revisited after considering the complete conversation, giving annotators a chance to change their initial aggressiveness rating after seeing the full context. The annotation was organized into 15 Google Forms, each containing batches of 10 threads with a different number of replies, and each form was annotated by five annotators. Threads with low agreement were removed, leaving 639 tweets and a Fleiss' Kappa of 0.6078. The annotators came from different countries, cultures, religions and age groups, providing diversity in the annotation process. The resulting corpus is particularly useful for this study because it contains both bystander-role labels and fine-grained cyberbullying labels in a session-based setting.

4.3 Bystander-Role Distribution

The dataset contains four bystander-role categories: instigator, defender, impartial and other. The paper reports that instigators and impartial participants are the most frequent roles across the thread categories. An instigator can reinforce the direction of the main post, while a defender can oppose the thread topic and support the victim. Impartial participants remain neutral or passive, whereas the other category covers replies that are unrelated to the thread topic. These distinctions provide social and conversational information that cannot be obtained from the main tweet alone.

4.4 Classifier Chain

Classifier Chain (CC) is a multi-label problem-transformation method in which classifiers are arranged sequentially. The prediction of an earlier label becomes an additional input feature for the next classifier. This differs from Binary Relevance (BR), where each label is predicted independently. The paper applies this idea to two dependent labels: the bystander role followed by the fine-grained cyberbullying class. [5]

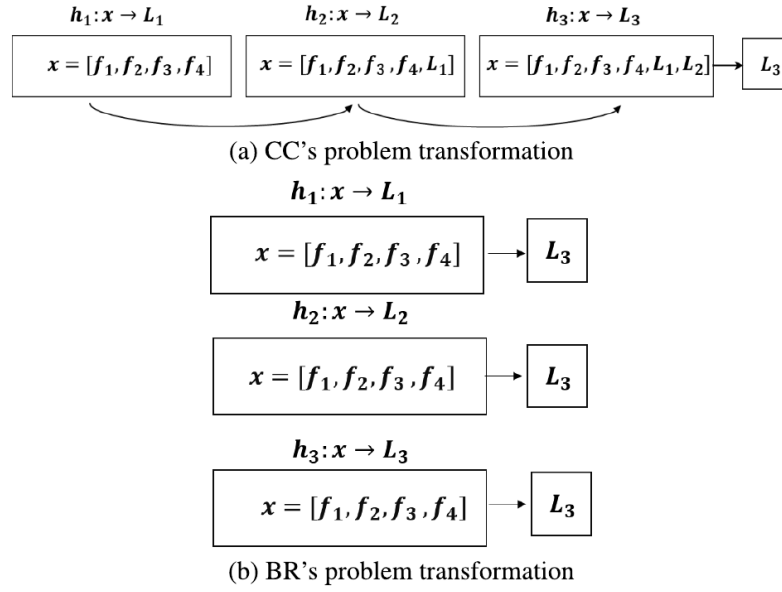


FIGURE 1. The comparison of BR and CC problem transformation methods for each classifier h_i , which is trained to predict L_i .

Figure 4.1: Comparison of Binary Relevance and Classifier Chain problem transformation.

4.4.1 Classifier Chain Formulation

For a feature set X , labels $Y = (y_1, \dots, y_n)$ and classifiers $F = (f_1, \dots, f_n)$, the process can be summarized as follows:

1. Train classifier f_i using the current feature set X and label y_i .
2. Predict the current label: $y'_i = f_i(X)$.
3. Append the prediction to the feature set: $X \leftarrow X \cup y'_i$.
4. Continue until all labels have been predicted.

In this paper, the chain has two stages. The first stage predicts bystander-role information, and the second stage uses that prediction along with the textual inputs to perform fine-grained cyberbullying classification.

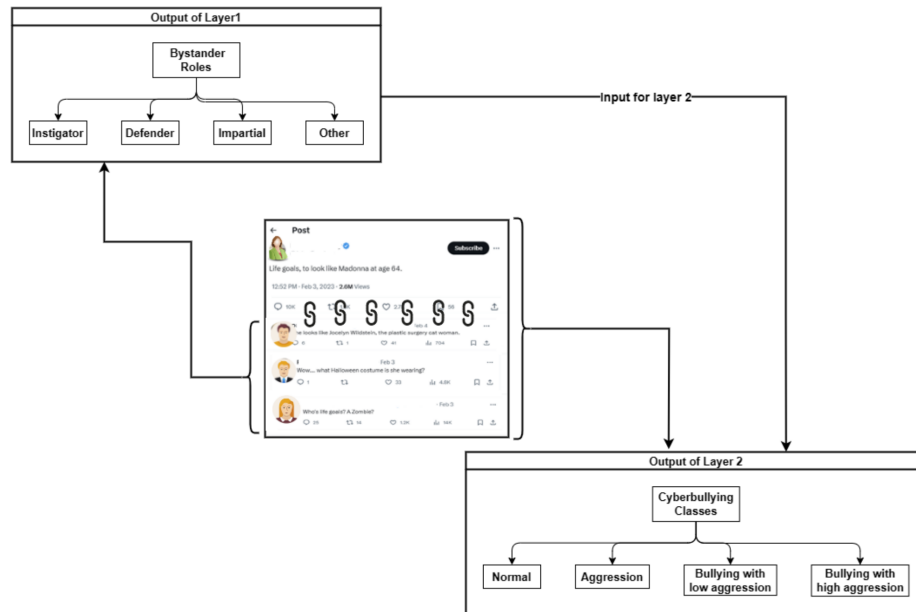


Figure 4.2: Two sequential layers of the chained fine-grained cyberbullying detection model.

4.5 Overall Methodology

The complete methodology proceeds from data collection and annotation to pre-processing, dataset splitting, and two-stage model training and evaluation. Layer 1 produces bystander-role information, while Layer 2 uses that information for fine-grained cyberbullying classification.

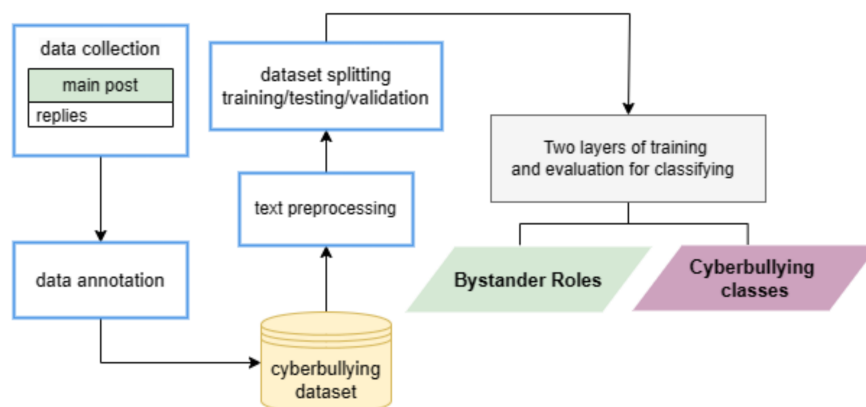


Figure 4.3: Overall methodology from data collection and annotation to two-stage classification.

4.6 Layer 1: Bystander Role Model using BERT

The first classifier is based on the pre-trained `bert-base-cased` model. The paper uses the cased version so that capitalization information is retained, which is useful because capitalization may carry meaning in social-media text. BERT contains 12 transformer layers with multiple attention heads and is fine-tuned for the specific bystander-role task. The implementation uses reply ID and tweet text, while the experiments also examine configurations that retain the user screen name. The tokenized sequence has a maximum length of 256 tokens and the selected batch size is 16.

Two dense layers are added above the BERT representation. The intermediate layer contains 512 units with a ReLU activation, followed by an output layer of 5 units with softmax activation as specified by the paper. The TensorFlow implementation is based on the Hugging Face Transformers library. AdamW is used as the optimization algorithm with a learning rate of 1×10^{-5} and weight decay of 1×10^{-6} . The reported fine-tuning configuration uses 5 epochs, taking approximately one hour per epoch, or about five hours in total. The authors also compare full fine-tuning with a frozen-BERT configuration and with RoBERTa. Full fine-tuning of BERT with the combined input configuration gives the best reported bystander-role performance, with weighted recall and weighted F1-score of 0.83. Freezing the lower BERT layers reduces performance, while the comparable RoBERTa configuration also performs below the fine-tuned BERT model.

4.7 BERT Bystander-Role Results

The paper evaluates several input configurations. The best reported configuration combines reply ID, tweet text and screen name and fully fine-tunes BERT. It achieves a weighted F1-score of 0.83.

Table 4.2: Bystander-role detection results for the reported BERT configurations.

Configuration	Precision	Recall	F1
Tweet text only	0.68	0.70	0.66
Tweet text + reply ID	0.77	0.77	0.75
Tweet text + screen name	0.80	0.74	0.73
All combined	0.82	0.83	0.83
Frozen BERT	0.71	0.70	0.69
RoBERTa	0.78	0.78	0.76

4.8 Layer 2: Fine-Grained Cyberbullying Model using LSTM

The second classifier is based on Long Short-Term Memory (LSTM), a recurrent neural architecture suitable for sequential data. In the reviewed model, three parallel LSTM branches process the reply ID, tweet text and predicted bystander-role information. Their learned representations are concatenated before the final classification layer. The paper describes LSTM as useful for learning dependencies across sequences through its memory cell and forget, input and output gates, helping it retain relevant information while reducing the vanishing-gradient difficulties associated with standard recurrent networks.

Before the recurrent layer, the text input is tokenized and represented using an embedding layer. The reported vocabulary size is 10,000 words and the embedding dimension is 128. The LSTM stage uses 64 hidden units in the selected configuration. Its branch outputs are concatenated and passed to a dense output layer with softmax activation so that the network produces class probabilities. The best reported configuration uses 64 hidden units, the Adam optimizer, 20 epochs, batch size 32, dropout of 0.5 and early stopping. The final output predicts three classes: bullying with high aggression, bullying with low aggression, and aggression without indication of bullying. The implementation also investigates 32-, 64- and 128-unit configurations, with 64 units giving the strongest overall evaluation according to the paper.

The authors further compare different training choices. Models trained for 20, 30, 50 and 100 epochs were evaluated, and 20 epochs produced the most promising results. Adding dropout with a rate of 0.5 gave a slight enhancement, while replacing LSTM with BiLSTM under the same configuration did not produce a notable improvement. Among Adam, SGD and RMSprop, SGD produced the lowest reported execution time of 30.58, but Adam produced the lowest validation loss of 0.30 and was selected as the more effective overall optimizer because it provided a better balance between loss and execution time.

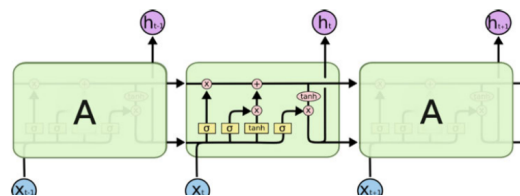


Figure 4.4: LSTM architecture used in the second stage.

4.9 Two-Stage Classification and Chaining

The two classifiers are connected so that the predicted bystander role from BERT is passed to the LSTM stage as additional information. The first layer therefore provides contextual role information, while the second layer uses the combined representation to determine the fine-grained cyberbullying class. This is the central contribution of the chained formulation.

4.10 Research Objectives and Contributions

The research is motivated by the observation that cyberbullying detection cannot always be treated as a simple decision about whether a single message is offensive. The authors formulate two main objectives. The first objective is to identify a useful feature associated with the role played by bystanders in a conversational thread. The second objective is to develop a fine-grained cyberbullying detection model that can use this bystander information to improve the semantic representation of the conversation and distinguish different levels of aggression. [1]

The contribution of the work can therefore be understood at three related levels. First, bystander behaviour is incorporated as an explicit feature rather than being ignored as background information. Second, the model introduces a chained architecture containing two sequential classification layers. Third, the proposed system is evaluated both with and without the bystander feature, allowing the authors to isolate its contribution to the final classification performance. This experimental design is particularly important because it turns the central research claim into a measurable comparison rather than relying only on an architectural argument.

4.11 Problem Formulation

The task addressed in the paper is a multi-label, multi-class classification problem. For each conversational instance, the input contains information related to a reply and its associated tweet text. The output contains two dependent labels: a bystander-role label and a fine-grained cyberbullying label. The bystander label has four categories, whereas the final cyberbullying label has three categories. Thus, the system must learn both the individual classification tasks and the relationship between them.

Let the input feature vector be represented by

$$X = [\text{reply_id}, \text{tweet text}],$$

and let the output be represented by

$$Y = [Y_1, Y_2],$$

where Y_1 denotes the bystander role and Y_2 denotes the fine-grained cyberbullying class. In an independent Binary Relevance formulation, Y_1 and Y_2 would be predicted separately. In the proposed chained formulation, the prediction $\hat{Y}_1$ obtained from the first classifier is appended to the information used by the second classifier. The second classifier therefore receives contextual information that is derived from the first prediction.

This formulation is useful because the labels are not completely independent. The way a participant responds to an aggressive conversation may provide evidence about the social dynamics of the thread, and those dynamics can in turn help distinguish bullying from aggression and differentiate levels of aggression. The classifier chain makes this dependency explicit within the model.

4.12 Data Collection and Construction of CYBY23

The dataset used in the experiments is CYBY23, a Twitter-based corpus designed to support fine-grained cyberbullying detection while incorporating bystander-role information. Twitter is particularly suitable for this setting because a user can publish an initial tweet and other users can respond to it, producing a thread containing the original post and associated replies. Such a thread provides more information than an isolated tweet because the replies reveal how other participants react to the original content.

The authors initially collected approximately 13,300 tweets over a twelve-month period from January 2022 to January 2023. From this larger collection, 150 tweet threads containing approximately 1,024 tweets were selected for detailed manual annotation. The reduction from the initial collection to a manually annotated subset reflects the practical difficulty of obtaining high-quality labels for conversational data. Annotation requires human judgement about aggression, context and participant roles, so applying detailed annotation to every collected tweet would be considerably more demanding.

The resulting dataset is valuable not simply because it contains text, but because it preserves information about the relationship between the original post and its replies. This makes it possible to examine the influence of bystanders on the final classification task. After the agreement-based filtering process, 639 tweets were retained for the experiments. The paper reports a Fleiss' Kappa value of 0.6078 for the annotation process, indicating a moderate level of agreement among the annotators.

4.13 Three-Level Annotation Strategy

The annotation procedure is an important part of the methodology because the final model depends on the quality of the labels. The paper describes annotation at three related levels. At the tweet level, the main tweet is considered and an aggressiveness score is assigned. At the reply level, responses are examined to determine the roles played by bystanders. At the thread level, annotators can reconsider their initial judgement after seeing the surrounding conversation.

The use of thread-level context is especially relevant to the problem. A short sentence may appear aggressive when viewed alone but may have a different interpretation after reading the preceding and following replies. Similarly, a reply may appear neutral without context but may be recognized as defending a participant once the topic of the thread is understood. The annotation procedure therefore attempts to reproduce the contextual reasoning required by the detection task.

The annotation was organized using 15 Google Forms, each containing batches of 10 threads with varying numbers of replies. Five annotators participated in the annotation. The use of multiple annotators helps reduce the effect of individual judgement, while the agreement filtering step removes threads for which the available annotations do not provide sufficient consistency. The paper also notes diversity among the annotators in terms of country, culture, religion and age, which is relevant because perceptions of aggressive language and social behaviour can vary across groups.

4.14 Label Structure of the Dataset

The final corpus contains two principal label dimensions. The first dimension represents the role of the bystander. The four categories are instigator, defender, impartial and other. The second dimension represents the nature and aggression level of the cyberbullying-related content. It contains bullying with high aggression, bullying with low aggression, and aggression without indication of bullying.

Table 4.3: Detailed label structure used in the CYBY23-based chained classification task.

Label dimension	Categories and interpretation
Bystander role	Instigator; Defender; Impartial; Other
Fine-grained class	Bullying with high aggression; Bullying with low aggression; Aggression without indication of bullying
Input context	Reply identifier and tweet text, with additional participant information considered during the BERT experiments

The distribution of the fine-grained classes is also important. Bullying with low aggression represents the largest class at 54.46%, aggression without bullying represents 33.93%, and bullying with high aggression represents 11.61%. The small proportion of the high-aggression class illustrates why overall accuracy alone can be misleading. A classifier could obtain a seemingly strong score by performing well on the dominant class while still making many errors on the minority class. For this reason, the paper emphasizes weighted evaluation metrics.

4.15 Data Preparation and Training Split

The experiments use a division of 60% of the data for training, 20% for validation and 20% for testing. The training set is used to learn the model parameters, the validation set is used to compare configurations and monitor generalization during development, and the test set is reserved for the final evaluation. Separating these roles helps prevent the final reported performance from being based on data directly used to choose the model configuration.

The paper’s methodology is also structured as a sequence rather than as a single end-to-end classification step. The first stage learns the bystander-role representation. Its output is then used by the second stage. This means that the data flow mirrors the conceptual relationship being investigated: first understand the participant’s role, then use that information while determining the severity and nature of the cyberbullying-related interaction.

4.16 Detailed BERT Architecture

BERT is used for the first layer because it can construct contextual representations of words by considering information from both directions of a sequence. The selected

model is **bert-base-cased**, which contains 12 transformer layers. The cased configuration preserves uppercase and lowercase distinctions. This is relevant to social-media text because capitalization can convey emphasis or emotional meaning.

The tokenization stage converts the input text into the token representation expected by BERT. The maximum sequence length is 256 tokens and the selected batch size is 16. The BERT representation is followed by two task-specific dense layers. The intermediate dense layer contains 512 units and uses ReLU activation. The output layer contains five units and uses softmax activation to produce probabilities for the five-class bystander-role prediction used in the implementation.

The model is fine-tuned rather than used only as a fixed feature extractor. AdamW is used as the optimizer with a learning rate of 1×10^{-5} and weight decay of 1×10^{-6} . The selected configuration is trained for five epochs, with approximately one hour reported per epoch. The paper also investigates what happens when the BERT layers are frozen. In that experiment, only the newly added layers are trained while the pre-trained BERT representation is kept fixed. The reduction in performance demonstrates that task-specific fine-tuning is beneficial for this challenging classification problem.

4.17 BERT Input Configuration Experiments

The first-stage experiments investigate which information is most useful for recognizing bystander roles. The authors compare configurations involving tweet text, reply identifiers and screen names. The strongest configuration retains the reply identifier and the screen name together with the tweet text and fully fine-tunes BERT. The paper explains that these identifiers help the model understand relationships between the main tweet and replies made by different participants in the same thread.

The BERT experiments also provide an important methodological lesson. Additional contextual information is useful only when it represents a meaningful relationship in the conversation. The improvement obtained from retaining reply and participant information suggests that the model is not relying solely on the lexical content of the message. Instead, the representation can include information about which reply belongs to which conversational context.

4.18 Frozen BERT and RoBERTa Comparison

Two additional experiments are used to examine whether the selected BERT configuration is justified. In the frozen-BERT experiment, the lower BERT layers are not retrained. This preserves the original pre-trained weights but reduces the ability

of the model to adapt its representations to the specific bystander-role classification task. The paper reports a decrease in performance under this configuration.

The authors also compare BERT with RoBERTa. RoBERTa is an enhanced transformer-based model with changes to the pre-training procedure, training data and hyperparameters. A comparable `roberta-base` configuration is evaluated. Despite the reputation of RoBERTa for strong classification performance, the reported experiments show that fine-tuned BERT is the stronger choice for this particular dataset and task, obtaining weighted recall and weighted F1-score of 0.83.

4.19 Detailed LSTM Architecture

The second stage uses LSTM because the task involves sequential text information. LSTM extends the recurrent neural-network idea with a memory cell and gating mechanisms. The gates regulate which information should be retained, which new information should be incorporated, and which information should be exposed as output. These mechanisms allow useful information to persist across a sequence and help address difficulties associated with long-term dependencies in ordinary recurrent networks.

In the proposed architecture, three parallel LSTM branches are used. One branch processes reply identifiers, another processes tweet text, and the third processes the predicted bystander-role information. The representations produced by these branches are concatenated before the final classification stage. This design allows the model to retain separate information streams before combining them into a unified representation.

For the text representation, the paper describes an embedding layer with a vocabulary size of 10,000 and an embedding dimension of 128. The experiments consider different hidden-layer sizes and identify 64 units as the strongest overall choice. The final dense layer uses softmax activation and produces probabilities for the three fine-grained cyberbullying classes.

4.20 LSTM Hyperparameter Experiments

The authors do not simply select a recurrent architecture without experimentation. They test several training configurations in order to identify an appropriate balance between classification quality and computational cost. Epoch counts of 20, 30, 50 and 100 are examined. The configuration using 20 epochs gives the most promising combination of precision, recall and F1-score while maintaining a low loss.

Regularization is also investigated. Early stopping is used with a patience of five epochs, meaning that training can stop when the validation error stops improving

and the best-performing weights are retained. A dropout rate of 0.5 is added to the LSTM configuration and produces a slight improvement. The paper also compares LSTM with BiLSTM under the same general configuration and reports no notable improvement from replacing LSTM with BiLSTM.

The optimizer experiment compares Adam, SGD and RMSprop. SGD provides the shortest reported execution time, 30.58, whereas Adam gives the lowest reported validation loss, 0.30. Adam is therefore considered the more effective overall choice because the objective is not merely to minimize execution time but to achieve strong predictive performance with acceptable computational cost.

4.21 Why the Chain Matters

The central difference between the proposed model and an independent pair of classifiers is the flow of information between the two tasks. If the bystander role were ignored, the second classifier would have to infer the social context only from the text and other available inputs. In the chained design, the first-stage prediction becomes an explicit feature for the second stage.

Consider a simplified conversational example. Suppose the main tweet contains aggressive language and a reply strongly supports the author. That reply may indicate an instigator role. In another thread, similar aggressive wording may be followed by a response that opposes the behaviour and supports the target, indicating a defender role. The textual similarity between the two cases does not necessarily imply identical social dynamics. The chained architecture is designed to preserve this distinction.

The approach also reflects the distinction between aggression and bullying. Aggressive language by itself does not always establish that an interaction is bullying. The surrounding thread, repetition, participant relationships and reactions may provide evidence that changes the interpretation. The proposed architecture therefore treats conversational context as part of the classification problem rather than as an optional external description.

4.22 Evaluation Metrics

Because the dataset is imbalanced, the paper evaluates the models using precision, recall and F1-score with weighted averaging. Precision measures the proportion of predicted positive instances that are correct, while recall measures the proportion of relevant instances that are successfully identified. F1-score combines precision and recall using their harmonic mean.

For a class c , precision and recall can be represented as

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}.$$

The corresponding F1-score is

$$F1_c = \frac{2P_cR_c}{P_c + R_c}.$$

For weighted evaluation, each class contribution is multiplied by its proportion in the dataset. If w_c denotes the fraction of samples belonging to class c , the weighted F1 can be written as

$$F1_{weighted} = \sum_c w_c F1_c.$$

This prevents the reported result from being interpreted without considering the class distribution.

The paper additionally evaluates reliability using confidence intervals. A 95% confidence interval provides an estimated range around the measured performance and gives additional information beyond a single point estimate. For the proposed model, the reported 95% confidence interval for weighted F1 is [0.78, 0.97].

Chapter 5

Results and Discussion

5.1 Experimental Setup

The experiments are reported using Google Colab Pro with a T4 GPU and a TensorFlow/Keras implementation. The dataset is divided into 60% training, 20% validation and 20% testing. Because the classes are imbalanced, weighted precision, weighted recall and weighted F1-score are used. The authors also report 95% confidence intervals for evaluation.

5.2 Baseline Evaluation

To isolate the contribution of bystander information, the paper constructs a baseline using the CY23 dataset, which contains individual tweets with fine-grained cyberbullying labels but no bystander-role or thread information. The same general classification objective is then compared with the proposed CYBY23 setting, where the complete thread and predicted bystander roles are available. This comparison is important because direct comparison with earlier studies is difficult when datasets and label definitions are different.

5.3 Impact of Bystander Roles

The main experiment compares the chained model with and without bystander-role information. The model with bystander roles achieves a weighted precision of 0.87, weighted recall of 0.90, weighted F1-score of 0.89 and test loss of 0.22. Removing the bystander-role information reduces the weighted F1-score to 0.71 and increases the loss to 0.59. [1]

Table 5.1: Impact of bystander-role information on final classification performance.

Approach	Precision	Recall	F1	Loss
With bystander roles	0.87	0.90	0.89	0.22
Without bystander roles	0.72	0.78	0.71	0.59

The increase from 0.71 to 0.89 corresponds to the reported 25.35% improvement over the no-bystander baseline. This is the strongest experimental evidence in the paper for the usefulness of explicitly modelling bystander roles.

5.4 Reliability of the Results

Beyond the point estimates of precision, recall and F1-score, the paper reports 95% confidence intervals (CI) computed using the t-distribution, in order to judge how trustworthy and precise the reported performance is.

Table 5.2: Confidence intervals for the proposed model and the baseline.

Approach	95% Confidence Interval (Weighted F1)
Proposed (with bystander roles)	[0.78, 0.97]
Baseline (without bystander roles)	[0.66, 0.75]

The two intervals do not overlap, which indicates that the improvement obtained by including bystander-role information is unlikely to be due to chance alone. This paired-interval check strengthens the earlier point comparison of 0.71 versus 0.89 weighted F1-score.

5.5 Comparison with Related Approaches

The paper compares its results with closely related studies. Because the datasets and tasks are not standardized across these studies, the comparison is indicative rather than a controlled benchmark. The reviewed model reports an F1-score of 89% while combining session-based data, bystander roles and fine-grained labels.

Table 5.3: Comparison with selected related cyberbullying detection approaches.

Study/Approach	Dataset/Setting	F1 (%)	Bystander Roles	Main characteristic
Ziems et al.	Twitter, session-based	77.9	No	Thread context; binary criteria
Wu and Tang	Chinese dialogues	61.76	No	Fine-grained severity
Jacobs et al.	ASKfm	60.04	Limited	Participant-role classification
Ge et al. (TGBully)	Instagram sessions	80.97	No	Temporal interaction modelling
Reviewed model	Twitter/CYBY23	89.0	Yes	Session-based; fine-grained

The comparison suggests that the reviewed model occupies a distinctive combination of session-level context, explicit bystander-role information and fine-grained cyberbullying severity. However, differences in datasets, label definitions and evaluation settings should be considered when interpreting the numerical values.

The paper also explains, study by study, why several of these related approaches report lower scores than the reviewed model:

- **Ziems et al.:** this study trains five independent logistic-regression classifiers, one per cyberbullying criterion, rather than a single chained model. Training the criteria independently is computationally expensive and discards the correlation between dependent cyberbullying attributes, which the authors link to the comparatively low reported F1-scores.
- **Jacobs et al.:** the ASKfm platform used for this study consists only of question-answer pairs rather than open threads, so only two bystander categories (harasser assistants and defenders) could be reliably captured, and the rare harasser-assistant class had to be merged with the harasser class. This narrower bystander scheme limits how much contextual information the model can exploit.
- **Ge et al. (TGBully):** this temporal graph-based approach requires linking each user's current interactions to their historical comments in order to model repetitive bullying behaviour, which is computationally costly. The paper also notes that, unlike traditional bullying, online abuse can remain permanently visible without needing to be repeated, which limits the benefit of this repetition-based modelling.
- **LSTM with TF-IDF/Bag-of-Words features:** approaches that rely on word-frequency features capture how often a word occurs but not its contextual or semantic meaning, which the paper identifies as the main reason their reported recognition accuracy is lower than that of the BERT-based bystander model.

5.6 Theoretical and Practical Implications

From a theoretical standpoint, social and educational research has long argued that bystanders shape how a bullying incident unfolds, but most automated cyberbullying detection work has continued to rely on binary, single-post classification. The paper attributes this gap mainly to the scarcity of session-based datasets that carry a bystander-role label alongside a fine-grained bullying label.

On the practical side, the paper highlights three concrete advancements of the proposed approach:

1. **Multi-label detection:** the model moves beyond single-label prediction to jointly support bystander-role and fine-grained cyberbullying labels.
2. **Capturing label dependency:** chained sequential classification lets the second-stage model use the first stage’s predicted label, making the relationship between the two labels explicit rather than implicit.
3. **An adapted chain classifier:** the paper presents an adapted version of the classifier-chain idea, since the standard implementation in libraries such as scikit-multilearn is designed for traditional machine-learning classifiers rather than deep-learning models such as BERT and LSTM.

Taken together, the authors position this chained formulation as a foundation that future studies can build on when incorporating bystander-role identification into fine-grained cyberbullying detection.

5.7 Discussion

The results support the main argument of the paper: bystander information is useful because it adds social context to the classification task. An instigator may reinforce an aggressive conversation, whereas a defender may counteract it; an impartial or unrelated reply provides different contextual evidence. By passing the predicted bystander information from the first classifier to the second classifier, the model can use this context when deciding the final severity class.

At the same time, the results should be interpreted in light of the size of CYBY23. The authors describe the dataset as a pilot corpus because it contains a relatively small number of reliably annotated tweets. The improvement in weighted F1 is therefore encouraging, but broader datasets and additional platforms are needed before generalization can be established.

5.8 Analysis of the Baseline Comparison

The baseline experiment is designed to answer a direct question: does the additional bystander-role information actually improve the final prediction, or is the improvement simply a consequence of changing the dataset? The authors address this by comparing the proposed CYBY23 setting with a baseline based on CY23, where the fine-grained cyberbullying label is available but the bystander-role and thread information are not incorporated.

The proposed model achieves a weighted precision of 0.87, weighted recall of 0.90 and weighted F1-score of 0.89, with a reported loss of 0.22. Without the bystander-role feature, the weighted precision is 0.72, weighted recall is 0.78 and weighted F1-score is 0.71, while the loss rises to 0.59. The weighted F1 therefore increases by 0.18 in absolute terms, and the paper reports this as a 25.35% improvement relative to the baseline.

The comparison is significant because it provides experimental evidence for the paper’s main hypothesis. The model does not merely use a larger or more complicated architecture; it specifically gains access to information describing the behaviour of participants in the conversation. The improvement indicates that this social information can help separate closely related classes, particularly when aggressive language alone is insufficient to determine whether an interaction should be classified as bullying.

5.9 Interpretation of the Confidence Interval

The paper reports a 95% confidence interval of [0.78, 0.97] for the weighted F1-score of the proposed model. This interval is useful because the final score of 0.89 should not be interpreted as an exact, universally fixed performance value. Instead, the interval describes the uncertainty associated with the estimate obtained from the available test data.

The authors use confidence intervals based on the t-distribution as an additional reliability check. The reported result is therefore supported by both a point estimate and an interval estimate. This is particularly relevant for a relatively small dataset, where relying on a single metric can give an incomplete picture of model reliability.

5.10 Experimental Findings from the Model Selection Process

The experiments collectively show that model performance depends on more than the choice of a neural-network name. Several design decisions contribute to the final system. For the first stage, retaining contextual participant information and fine-tuning BERT are beneficial. For the second stage, an LSTM with an appropriate hidden-layer size, dropout and early stopping provides the selected configuration. Optimizer choice also affects the trade-off between validation loss and execution time.

The hidden-layer experiment is especially useful for understanding the practical trade-off. A smaller network may reduce computational requirements, but the

paper reports that 64 hidden units provide the strongest overall evaluation among the tested 32-, 64- and 128-unit configurations. The result illustrates that increasing model size does not automatically produce a proportional increase in predictive quality.

Similarly, the optimizer comparison shows that the fastest configuration is not necessarily the most effective configuration. SGD achieves the lowest execution time reported in the experiment, while Adam obtains the lowest validation loss. The final selection therefore considers both computational efficiency and predictive behaviour.

5.11 Error Propagation in the Chained Architecture

A chained architecture introduces an important dependency: the second classifier receives information generated by the first classifier. This creates a potential error-propagation pathway. If the BERT stage assigns an incorrect bystander role, the LSTM stage may receive misleading contextual information and make a corresponding error in the final classification.

This issue is not a minor implementation detail; it is an inherent property of sequential classification. The paper therefore treats the fixed chain order as both a strength and a limitation. The strength is that the model explicitly captures the relationship between the labels. The limitation is that errors from an earlier stage can influence later decisions.

The authors retain a fixed order because their theoretical motivation places bystander-role information before the final cyberbullying-severity decision. They argue that bystander replies influence the dynamics of the bullying event, making the role prediction a meaningful first step. Future improvements can attempt to preserve this dependency while reducing the effect of incorrect first-stage predictions.

5.12 Comparison with Earlier Detection Strategies

The reviewed approach differs from several earlier families of cyberbullying detection systems. Traditional text classifiers often operate on isolated posts and use word-frequency features such as Bag-of-Words or TF-IDF. These representations are useful for measuring lexical occurrence, but they do not directly model the contextual relationship between words and participants in a conversation.

Other approaches incorporate session or thread information but may still use binary labels. Session-based methods are closer to the conversational setting of CYBY23 because they allow the complete interaction to be considered. However,

the reviewed work goes a step further by assigning explicit bystander roles and then using those roles in a chained fine-grained classifier.

Some participant-role studies classify users into categories such as harasser, victim and bystander. Although such work recognizes that participant identity matters, the reviewed paper focuses specifically on the role of bystanders within the conversation and connects that role to the severity of cyberbullying. This creates a direct link between social interaction and the final fine-grained label.

The paper also discusses severity-oriented approaches such as hierarchical squashing-attention networks. These models demonstrate that cyberbullying can be represented using multiple severity levels instead of a binary decision. The limitation, from the perspective of the reviewed work, is that severity can still be interpreted without explicitly modelling the influence of bystanders. The proposed chain attempts to combine the two ideas.

5.13 Practical Significance

The practical importance of the proposed architecture lies in its ability to reduce a common ambiguity in automated moderation: aggressive language does not always mean cyberbullying. A system that examines only the words may flag jokes, isolated arguments or criticism as bullying, while a system that understands conversational context can potentially distinguish these cases more effectively.

Bystander responses can provide several forms of evidence. A defending response can indicate opposition to the aggressive behaviour. An instigating response can reinforce the behaviour. An impartial response provides little direct support for either side, while an unrelated response may indicate that the participant is not engaged with the bullying topic. By encoding these roles, the system introduces a social dimension into a primarily linguistic classification problem.

The approach can therefore be viewed as a step toward context-aware moderation systems. In a practical deployment, such a system could act as a decision-support component that prioritizes conversations for human review, rather than automatically making irreversible moderation decisions. The paper itself, however, evaluates the research model on the CYBY23 dataset and does not establish deployment-level performance across live social-media platforms.

5.14 Overall Interpretation of the Results

The central experimental result is consistent across the main comparisons: incorporating bystander roles improves the final fine-grained classification. The proposed system reaches a weighted F1-score of 0.89, compared with 0.71 for the corresponding

no-bystander baseline. The reported confidence interval and the additional model-selection experiments provide supporting evidence that the improvement is not simply an artifact of one arbitrary configuration.

At the same time, the paper is careful about the limits of direct comparison with previous work. There is no standardized session-based dataset containing both bystander-role labels and fine-grained cyberbullying labels. Consequently, the authors compare their results with the most similar published approaches rather than claiming a perfectly controlled benchmark. This distinction is important when interpreting the reported 89% F1-score.

Chapter 6

Limitations and Future Scope

6.1 Limitations

The paper identifies several limitations in the proposed chained architecture and dataset:

1. **Fixed chain order:** the second-stage classifier depends on the prediction made by the first stage, so an error in bystander-role prediction can propagate forward and reduce the accuracy and reliability of the overall model. Most earlier chained-classifier studies address this by searching for an optimal chain order, but this paper keeps a fixed order, since the authors’ theoretical grounding is that bystander replies significantly influence the dynamics of cyberbullying and should therefore be predicted first.
2. **Small annotated dataset:** only 639 tweets remained after removing low-agreement annotations, limiting the scale of training and evaluation.
3. **Single dataset setting:** all experimental stages are limited to the CYBY23 dataset, since it is the only available dataset that labels both bystander roles and fine-grained bullying severity; this restricts the ability to verify performance across different platforms and annotation schemes.
4. **Role scope:** the model focuses on bystander roles, tagging participants as instigator, defender, impartial, or other, and does not extend to explicitly classifying participants as bully or victim in the final task.

6.2 Future Scope

The paper suggests several concrete directions for future work:

- **Reduce error propagation:** explore mitigation mechanisms such as ensemble methods, confidence thresholds, uncertainty estimation, and feedback loops between the two chain stages.
- **Expand the annotated corpus:** collaborate with domain experts to build a larger, more extensively expert-labelled dataset, since qualified annotators can improve label quality and, in turn, model performance.

- **Generalize across platforms:** extend the session-based approach to datasets collected from other social-media platforms, such as Facebook and YouTube, to test whether the same relationship between bystander roles and bullying severity still holds.
- **Address class imbalance:** rebalance the dataset, either by increasing the size of the smallest classes (such as bullying with high aggression) or by applying data augmentation techniques, since real-world cyberbullying data is inherently imbalanced.

These directions would help determine whether the observed relationship between bystander roles and cyberbullying severity remains consistent across larger and more diverse conversational datasets.

Chapter 7

Conclusion

This seminar reviewed a chained deep learning approach for fine-grained cyberbullying detection that explicitly incorporates the behaviour of bystanders in a conversation thread. The central idea is to move beyond binary classification of an isolated post and instead model the interaction between bystander roles and cyberbullying severity.

The model uses a fine-tuned BERT classifier as the first stage to identify bystander-role information and an LSTM-based classifier as the second stage to determine the final fine-grained cyberbullying class. A classifier-chain formulation connects the two stages by passing the first-stage prediction to the second stage as additional information. [1]

The reported experiments on CYBY23 show a weighted F1-score of 0.89 when bystander roles are included, compared with 0.71 for the no-bystander baseline. This 25.35% improvement supports the paper’s conclusion that bystander behaviour can provide valuable contextual information for fine-grained cyberbullying detection. [1]

Overall, the work demonstrates how conversational context and dependent label modelling can be combined in a practical deep learning architecture. Its main limitation is the relatively small evaluation dataset, but the approach provides a useful direction for future cyberbullying detection systems that need to understand not only what is said, but also how other participants respond. The study’s main experimental finding is that the additional social context provided by bystander roles can materially improve fine-grained classification. The reported 95% confidence interval for the proposed model’s weighted F1-score is [0.78, 0.97], further supporting the reliability of the observed performance on the available test data. At the same time, the authors emphasize that the absence of other session-based datasets with both bystander and fine-grained labels prevents a standardized cross-dataset evaluation.

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